Aarav Verma

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NOBE, Project Manager UAV@Illinois, Build Team Engineer Incoming L3Harris Intern

CAD AND HARDWARE PROJECTS



Haylon Technologies Interceptor Drone

June 2026 to Present

NOBE for Haylon Technologies, Project Manager

Led a six-developer team through a 12-week hardware and software roadmap: wrote every weekly deliverable and rubric, reviewed all CAD and code, and ran the design competition that chose the final airframe. Streamlined fuselage with four split motor nacelles, screw-on winglets, an electronics stack that slides out with the nose cap in one piece, and a forward AKK BA3 camera mount on M3 heat-set inserts.

Autodesk Fusion, 3D Printing, ESP32, Project Management



Full Drone and Computer System

Feb 2026 to May 2026

NOBE, Canary Drone Project, Engineering Lead

Designed a modular drone airframe in Fusion 360 with a Jetson housing and tool-free foldable leg motors on 4 snap-lock hinges. FEA under 12N thrust load confirmed FOS above 2.0 across a 50+ component BOM. Presented at the Cozad New Venture Challenge against 300+ teams for $500,000.

Fusion 360, FEA, Avionics Integration



Toroidal Propeller Design

Feb 2026 to May 2026

NOBE, Canary Drone Project, Engineering Lead

Redesigned a 2-blade propeller to a toroidal configuration in Fusion 360. Enclosing the blade-tip vortex path raised thrust efficiency about 11% at 4 to 6k RPM and cut acoustic signature below 60 dB at 1m hover, with FOS above 2.0.

Fusion 360, FEA, Aerodynamics



Swingarm Assembly

Aug 2025 to Jan 2026

Ghost Electric Motorcycles, Drivetrain Engineer

Reverse-engineered a rear suspension swingarm from the 2010 Kawasaki ZX-6R in SolidWorks to replicate OEM anti-squat ratio and pivot geometry, then validated stiffness under braking, acceleration, and cornering loads.

SolidWorks, ANSYS FEA, MATLAB



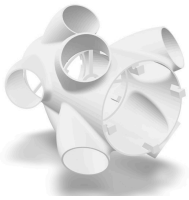
GHOST Drivetrain Frame

Aug 2025 to Jan 2026

Ghost Electric Motorcycles, Drivetrain Engineer

Engineered a load-bearing drivetrain frame in SolidWorks to align motor, top rod, and swingarm to ±0.05 mm. ANSYS and MATLAB confirmed stiffness under peak torque, chain tension, and rider weight across all loading conditions.

SolidWorks, ANSYS FEA, MATLAB



Revolving Modular Hydroponic Tower 2024

UIUC Residence Hall Idea Fair, Designer and Builder

Designed a revolving modular hydroponic tower in Fusion 360, building the stacking system around FDM print tolerances for every interlocking joint. Printed and assembled the full prototype in-house.

Fusion 360, 3D Printing, DFM

SOFTWARE PROJECTS



Haylon Intercept Sim github.com/Verma0118/haylon-intercept-sim

Browser simulation of the interceptor's control idea: detect, stabilize, track, and guide. Throw a target with straight, weaving, or erratic evasion and watch the guidance law chase it in 3D. Built before any CAD or hardware existed so the control logic was proven first.

JavaScript, Three.js, Vite, Guidance, State Machine

Volant github.com/Verma0118/volant

B2B SaaS fleet operations platform for drone and eVTOL operators. Real-time telemetry, mission dispatch, and fleet management.

React, TypeScript, Node.js, PostgreSQL

EmailCRM Autopilot github.com/Verma0118/email-autopilot

Daily outreach loop that syncs my inbox and drafts replies and cold emails into an approval queue. Python owns Gmail and the filesystem, Claude handles judgment, and nothing sends without my approval.

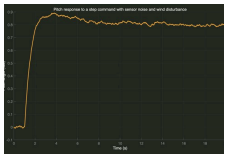
Python, Gmail API, React, Claude

TVC Simulator github.com/Verma0118/TVC_Simulator_V1

Thrust Vector Control simulator in Python modeling gimbal-actuated rocket engine dynamics for attitude control during powered flight.

Python, Simulation, Controls

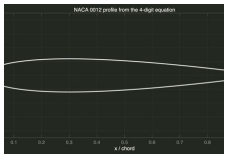
SIMULATION AND ANALYSIS



Cascaded Flight Controller for Pitch, Roll, and Yaw 2026

Built a multi-axis PID attitude controller in Simulink from first principles, deriving plant transfer functions from Newton's second law. Restructured pitch into a cascaded inner-rate, outer-angle architecture matching Betaflight and PX4, fixed derivative kick and noise amplification at the root cause, and validated disturbance rejection against wind gusts and sensor noise.

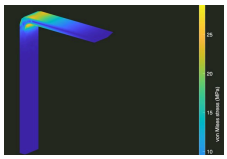
Simulink, PID Control, MATLAB



CAD-to-FEA Pipeline for a NACA 0012 Airfoil 2026

Generated NACA 0012 geometry in MATLAB, built the solid in Fusion 360, and ran stress analysis in MATLAB's FEA solver. A mesh convergence study caught a stress singularity at an unfilleted corner, fixed through CAD redesign and re-validated.

MATLAB, FEA, Fusion 360, Mesh Convergence



Hand-Validated Cantilever Bracket FEA 2026

Hand-calculated deflection and bending stress before simulating, then traced a 10x discrepancy between hand calc and FEA to a wrong assumed load-face area and fixed it by querying the true area from the mesh. Reconciled the remaining gap with bend-fillet compliance and hole stress concentration.

MATLAB, FEA, Mechanics of Materials

EXPERIENCE AND SKILLS

Organizations

- NOBE, Project Manager
- UAV@Illinois, Build Team Engineer

Previous Work

- Ghost Electric Motorcycles, Drivetrain Engineer
- Illinois Space Society, Space Shot Structural Engineer
- Tarform, Mechanical Design Engineer Intern (Part-Time)

CAD and Simulation

- NX Siemens, SolidWorks, Fusion
- CATIA V5, ANSYS
- MATLAB, Simulink

Software

- Python, TypeScript, React
- C++, Git, ROS

Manufacturing

- FDM 3D Printing, CNC Machining
- Laser Cutting, Sheet Metal